Math 241, F14
Exam 2

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points)
(a) Use Chain Rule to find $\partial z/\partial s$ and $\partial z/\partial t$ in terms of s and t when $z = \ln(2x+3y)$, $x = te^s$, $y = e^{st}$.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} = \frac{2}{2x+3y}\cdot(te^s) + \frac{3}{2x+3y}(te^{st})$$

$$= \frac{2te^s}{2te^s+3e^{st}} + \frac{3te^{st}}{2te^s+3e^{st}}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} = \frac{2}{2x+3y}(e^s) + \frac{3}{2x+3y}(se^{st})$$

$$= \frac{2e^s}{2te^s+3e^{st}} + \frac{3se^{st}}{2te^s+3e^{st}}$$

(b) Given $f(x,y) = x^2 - 4xy + 2y^2$, $\mathbf{p} = (-1,2)$ and $\mathbf{a} = \mathbf{i} - \sqrt{3}\mathbf{j}$, find the directional derivative of f at the point $\mathbf{p}$ in the direction of $\mathbf{a}$.

$$f_x = 2x - 4y, \quad f_y = -4x + 4y$$

$$\Rightarrow \nabla f(-1,2) = \langle 2\cdot(-1) - 4\cdot 2, -4\cdot(-1) + 4\cdot 2\rangle$$

$$= \langle -10, 12\rangle$$

$$a = \langle 1, -\sqrt{3}\rangle \Rightarrow u = \frac{a}{|a|} = \frac{\langle 1, -\sqrt{3}\rangle}{\sqrt{1^2+(-\sqrt{3})^2}} = \langle \frac{1}{2}, -\frac{\sqrt{3}}{2}\rangle$$

$$\Rightarrow D_u f(-1,2) = \langle -10, 12\rangle \cdot \langle \frac{1}{2}, -\frac{\sqrt{3}}{2}\rangle$$

$$= -5 - 6\sqrt{3}$$

Problem 2. (22 points)
(a) Find all critical points of $f(x,y) = x^2y - 6y^2 - 3x^2$ and then determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point.

$$\begin{cases} f_x = 2xy - 6x = 0 \Rightarrow 2x(y-3) = 0 \\ f_y = x^2 - 12y = 0 \Rightarrow y = \frac{x^2}{12} \end{cases}$$

$\Rightarrow 2x\left(\frac{x^2}{12} - 3\right) = 0$

$x(x^2-36) = 0$

$x^2(x+6)(x-6) = 0$

$x = 0, -6, 6$

critical points: $(0,0)$, $(-6,3)$, $(6,3)$

$f_{xx} = 2y - 6$, $f_{yy} = -12$, $f_{xy} = 2x$.

$D = f_{xx}f_{yy} - f_{xy}^2$

For $(0,0)$, $f_{xx}(0,0) = -6 < 0$, $D(0,0) = (-6)(-12) - 0 = 72 > 0$

$\Rightarrow (0,0)$ gives a local maximum

For $(-6,3)$, $D(-6,3) = (0)\cdot(-12) - (-12)^2 = -144 < 0$

$\Rightarrow (-6,3)$ is a saddle point

For $(6,3)$, $D(6,3) = (0)\cdot(-12) - (12)^2 = -144 < 0$

$\Rightarrow (6,3)$ is a saddle point.

(b) Use the method of **Lagrange multipliers** to find the maximum and minimum value of the function $f(x,y) = x + 3y$ subject to the constraint $x^2 + y^2 = 40$.

$\nabla f = \langle 1, 3 \rangle$, $\nabla g = \langle 2x, 2y \rangle$ where $g(x,y) = x^2 + y^2$

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 40 \end{cases} \Rightarrow \begin{cases} 1 = 2\lambda x \Rightarrow x = \frac{1}{2\lambda} \\ 3 = 2\lambda y \Rightarrow y = \frac{3}{2\lambda} \\ x^2 + y^2 = 40 \end{cases}$$

$\Rightarrow \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{3}{2\lambda}\right)^2 = 40$

$\frac{10}{4\lambda^2} = 40 \Rightarrow \lambda^2 = \frac{10}{160} = \frac{1}{16}$

$\Rightarrow \lambda = \frac{1}{4}, -\frac{1}{4}$

$\lambda = \frac{1}{4} \Rightarrow x = 2, y = 6 \Rightarrow f(2,6) = 20$

$\lambda = -\frac{1}{4} \Rightarrow x = -2, y = -6 \Rightarrow f(-2,-6) = -20$

$\Rightarrow$ The maximum value : 20

minimum value : -20

Problem 3. (20 points) Evaluate the following iterated integrals.

(a) $\displaystyle\int_0^1 \int_0^x \frac{2y}{x^3+1}\, dydx$

$$= \int_0^1 \frac{1}{x^3+1}\left[y^2\right]_{y=0}^{y=x} dx$$

$$= \int_0^1 \frac{x^2}{x^3+1}\, dx$$

$$= \left[\frac{1}{3}\ln(x^3+1)\right]_{x=0}^{x=1}$$

$$= \frac{1}{3}\ln 2 - \frac{1}{3}\ln 1$$

$$= \frac{1}{3}\ln 2$$

(b) $\displaystyle\int_0^2 \int_z^1 \int_0^{\sqrt{x/z}} 2xyz\, dydxdz$

$$= \int_0^2 \int_z^1 xz\left[y^2\right]_{y=0}^{y=\sqrt{x/z}} dxdz$$

$$= \int_0^2 \int_z^1 xz \cdot \frac{x}{z}\, dxdz$$

$$= \int_0^2 \int_z^1 x^2 dxdz$$

$$= \int_0^2 \left[\frac{x^3}{3}\right]_{x=z}^{x=1} dx$$

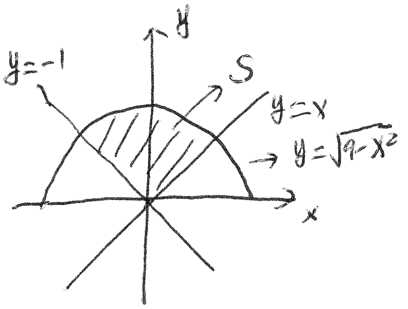
$$= \int_0^2 \frac{1-z^3}{3}\, dx$$

$$= \left[\frac{z}{3} - \frac{z^4}{12}\right]_{z=0}^{z=2}$$

$$= \frac{2}{3} - \frac{16}{12} = -\frac{2}{3}$$

Problem 4. (20 points)

(a) A lamina covering the region S in the plane is bounded by the curves $y = x$, $y = -x$ and $y = \sqrt{9 - x^2}$. Assume it has a density $f(x, y) = \sqrt{x^2 + y^2}$. Find the mass of the lamina by evaluating $\iint_S f(x, y)\, dA$ in the **polar coordinates**.



$$S = \{(r,\theta) \mid 0 \le r \le 3,\ \tfrac{\pi}{4} \le \theta \le \tfrac{3\pi}{4}\}$$

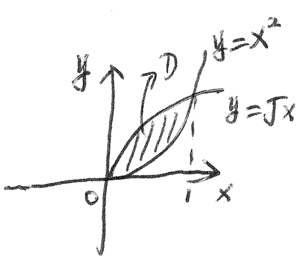
$$f(x,y) = \sqrt{x^2+y^2} = r$$

$$\iint_S \sqrt{x^2+y^2}\, dA = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^3 r \cdot r\, dr\, d\theta = \int_{\pi/4}^{3\pi/4} \left[\frac{r^3}{3}\right]_{r=0}^{r=3}$$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 9\, d\theta$$

$$= 9\left(\frac{3\pi}{4} - \frac{\pi}{4}\right) = 9 \times \frac{\pi}{2} = \frac{9}{2}\pi$$

(b) The solid E lies under the surface $z = 2xy$ and above the region D in the xy-plane enclosed by $y = x^2$ and $y = \sqrt{x}$. Find the volume of E by computing $\iiint_E dV$.



$$\begin{cases} y = x^2 \\ y = \sqrt{x} \end{cases} \Rightarrow \begin{matrix} x=0, & x=1 \\ \Downarrow & \Downarrow \\ y=0 & y=1 \end{matrix}$$

$$D = \{(x,y) \mid 0 \le x \le 1,\ x^2 \le y \le \sqrt{x}\}.$$

$$\iiint_E dV = \int_0^1 \int_{x^2}^{\sqrt{x}} \int_0^{2xy} dz\, dy\, dx$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} 2xy\, dy\, dx = \int_0^1 \left[xy^2\right]_{y=x^2}^{y=\sqrt{x}} dx$$

$$= \int_0^1 x^2 - x^5\, dx$$

$$= \left[\frac{x^3}{3} - \frac{x^6}{6}\right]_{x=0}^{x=1} = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$$

Problem 5. (20 points)

(a) Evaluate $\iiint_E \sqrt{x^2+y^2+z^2}\, dV$ where E is the solid enclosed by $x^2+y^2+z^2=4$, $y \geq 0$, and $z \geq 0$ using the triple integral in the **spherical coordinates**.

$$E=\{(\rho,\theta,\phi)\mid 0\leq\rho\leq 2,\ 0\leq\theta\leq\pi,\ 0\leq\phi\leq\tfrac{\pi}{2}\}$$

$$\iiint_E \sqrt{x^2+y^2+z^2}\,dV=\int_0^2\int_0^{\pi}\int_0^{\frac{\pi}{2}} \rho\cdot\rho^2\sin\phi\, d\phi\, d\theta\, d\rho$$

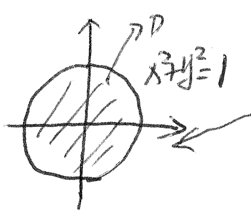
$$=\int_0^2\int_0^{\pi} \rho^3\left[-\cos\phi\right]_{\phi=0}^{\phi=\frac{\pi}{2}} d\theta\, d\rho$$

$$=\int_0^2\int_0^{\pi} \rho^3 d\theta\, d\rho$$

$$=\int_0^2 \pi\rho^3 d\rho=\left[\frac{\pi\rho^4}{4}\right]_{\rho=0}^{\rho=2}$$

$$=4\pi$$

(b) Using the **cylindrical coordinates** to compute the triple integral $\iiint_E z\, dV$ where E is the solid bounded above by the paraboloid $z = 1 - x^2 - y^2$ and below by the plane $z = 0$.



$$\begin{cases} z=0 \\ z=1-x^2-y^2 \end{cases} \Rightarrow x^2+y^2=1$$

$$\Rightarrow E=\{(r,\theta,z)\mid \overbrace{0\leq r\leq 1,\ 0\leq\theta\leq 2\pi}^{D},\ 0\leq z\leq 1-r^2\}$$

$$\iiint_E z\,dV=\int_0^{2\pi}\int_0^1\int_0^{1-r^2} z r\, dz\, dr\, d\theta$$

$$=\int_0^{2\pi}\int_0^1 \left[\frac{z^2}{2}\right]_{z=0}^{z=1-r^2} r\, dr\, d\theta$$

$$=\frac{1}{2}\int_0^{2\pi}\int_0^1 r(1-r^2)^2 dr\, d\theta=\frac{1}{2}\int_0^{2\pi}\int_0^1 r-2r^3+r^5 dr\, d\theta$$

$$=\frac{1}{2}\int_0^{2\pi}\left[\frac{r^2}{2}-\frac{1}{2}r^4+\frac{1}{6}r^6\right]_{r=0}^{r=1} d\theta$$

$$=\frac{1}{2}\int_0^{2\pi}\frac{1}{2}-\frac{1}{2}+\frac{1}{6}\, d\theta=\frac{1}{2}\int_0^{2\pi}\frac{1}{6}d\theta$$

$$=\frac{\pi}{6}$$